

Lab 8: Hands-On Remote Login Demonstration across Protocols (VPN vs RDP vs SSH vs VDI)

Lab 8: Protocol Comparison Lab: Conduct hands-on login demonstrations across VPN, RDP, SSH, and VDI platforms, evaluating performance and security.

Introduction

Different remote access technologies serve distinct enterprise roles. This lab provides hands-on login demonstrations across VPN, RDP, SSH, and VDI sessions to evaluate traffic overhead, authentication mechanics, and operational usability.

Objectives

Aim: Execute and record interactive remote logins using VPN, RDP, SSH, and VDI, documenting hands-on session characteristics.

- Perform live remote login sessions using OpenVPN, RDP, SSH, and VDI tools.
- Capture packet samples for each session type using Wireshark.
- Analyze protocol overhead, encryption indicators, and user experience differences.

Requirements

- Integrated lab environment with OpenVPN server, RDP host, SSH server, and VDI broker.
- Wireshark packet analyzer.

Expected Output

- Comparative log document recording login sequence and packet capture analysis for all 4 protocols.
- Protocol traffic identification matrix.

Step by Step Procedure(s)

Step-1: Start Network Capture: Launch Wireshark on client interface.

```
C:\Users\Kiran>ssh kiran@192.168.233.149
kiran@192.168.233.149's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-14-generic x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro
```

Expanded Security Maintenance for Applications is not enabled.

378 updates can be applied immediately.
303 of these updates are standard security updates.
To see these additional updates run: `apt list --upgradable`

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

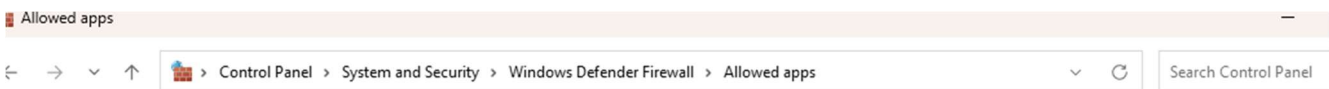
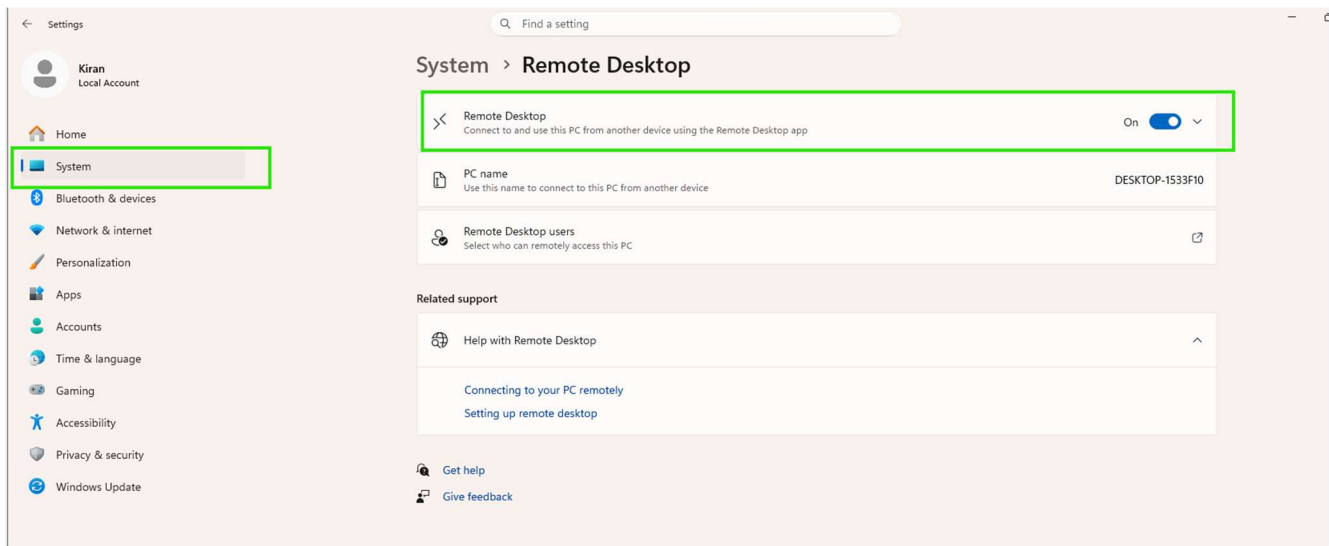
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

```
kiran@kiran-VMware:~$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libncurses6 libncursesw6 libtinfo6 ncurses-base ncurses-bin ncurses-term
  openssh-client openssh-sftp-server ssh-import-id
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
The following packages will be upgraded:
  libncurses6 libncursesw6 libtinfo6 ncurses-base ncurses-bin openssh-client
6 upgraded, 4 newly installed, 0 to remove and 370 not upgraded.
```

```
kiran@kiran-VMware:~$ sudo systemctl start ssh
kiran@kiran-VMware:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
kiran@kiran-VMware:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-03 18:29:21 IST; 49min ago
 TriggeredBy: ● ssh.socket
    Docs: man:sshd(8)
          man:sshd_config(5)
   Main PID: 6087 (sshd)
     Tasks: 1 (limit: 4544)
    Memory: 3.2M (peak: 4.2M)
       CPU: 68ms
    CGroup: /system.slice/ssh.service
            └─6087 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 03 18:51:44 kiran-VMware sshd[6166]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh rus
Aug 03 18:51:46 kiran-VMware sshd[6166]: Failed password for invalid user kiran-VMware from 192.168.233.153 port 62467
Aug 03 18:51:51 kiran-VMware sshd[6166]: pam_unix(sshd:auth): check pass; user unknown
Aug 03 18:51:54 kiran-VMware sshd[6166]: Failed password for invalid user kiran-VMware from 192.168.233.153 port 62467
Aug 03 18:52:21 kiran-VMware sshd[6166]: pam_unix(sshd:auth): check pass; user unknown
Aug 03 18:52:23 kiran-VMware sshd[6166]: Failed password for invalid user kiran-VMware from 192.168.233.153 port 62467
Aug 03 18:52:23 kiran-VMware sshd[6166]: Connection reset by invalid user kiran-VMware 192.168.233.153 port 62467 [prea
Aug 03 18:52:23 kiran-VMware sshd[6166]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost
Aug 03 18:55:01 kiran-VMware sshd[6169]: Accepted password for kiran from 192.168.233.153 port 62474 ssh2
Aug 03 18:55:01 kiran-VMware sshd[6169]: pam_unix(sshd:session): session opened for user kiran(uid=1000) by kiran(uid=0
lines 1-23/23 (END)
```

```
kiran@kiran-VMware:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:bd:b6:a0 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.22.144/24 brd 192.168.22.255 scope global dynamic noprefixroute ens33
        valid_lft 1396sec preferred_lft 1396sec
    inet6 fe80::24ad:f4e6:38c1:6481/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:bd:b6:aa brd ff:ff:ff:ff:ff:ff
    altname enp2s2
    inet 192.168.233.149/24 brd 192.168.233.255 scope global dynamic noprefixroute ens34
        valid_lft 1396sec preferred_lft 1396sec
    inet6 fe80::20c:29ff:febd:b6aa/64 scope link
```

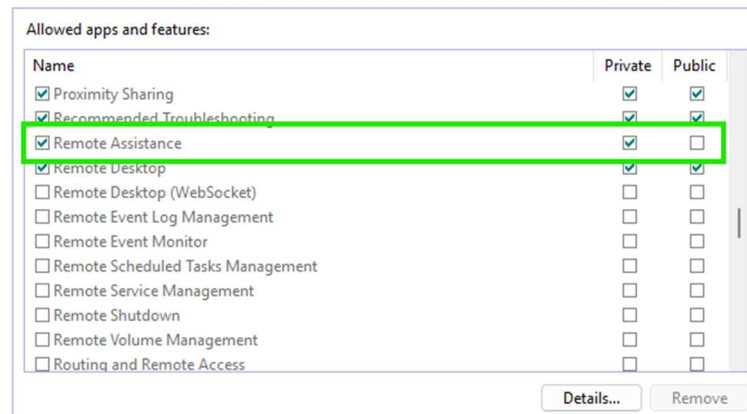


Allow apps to communicate through Windows Defender Firewall

To add, change, or remove allowed apps and ports, click Change settings.

What are the risks of allowing an app to communicate?

[Change settings](#)



```

PS C:\Users\Kiran> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::8459:b4b7:7028:d93b%13
    IPv4 Address. . . . . : 192.168.22.139
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.22.2

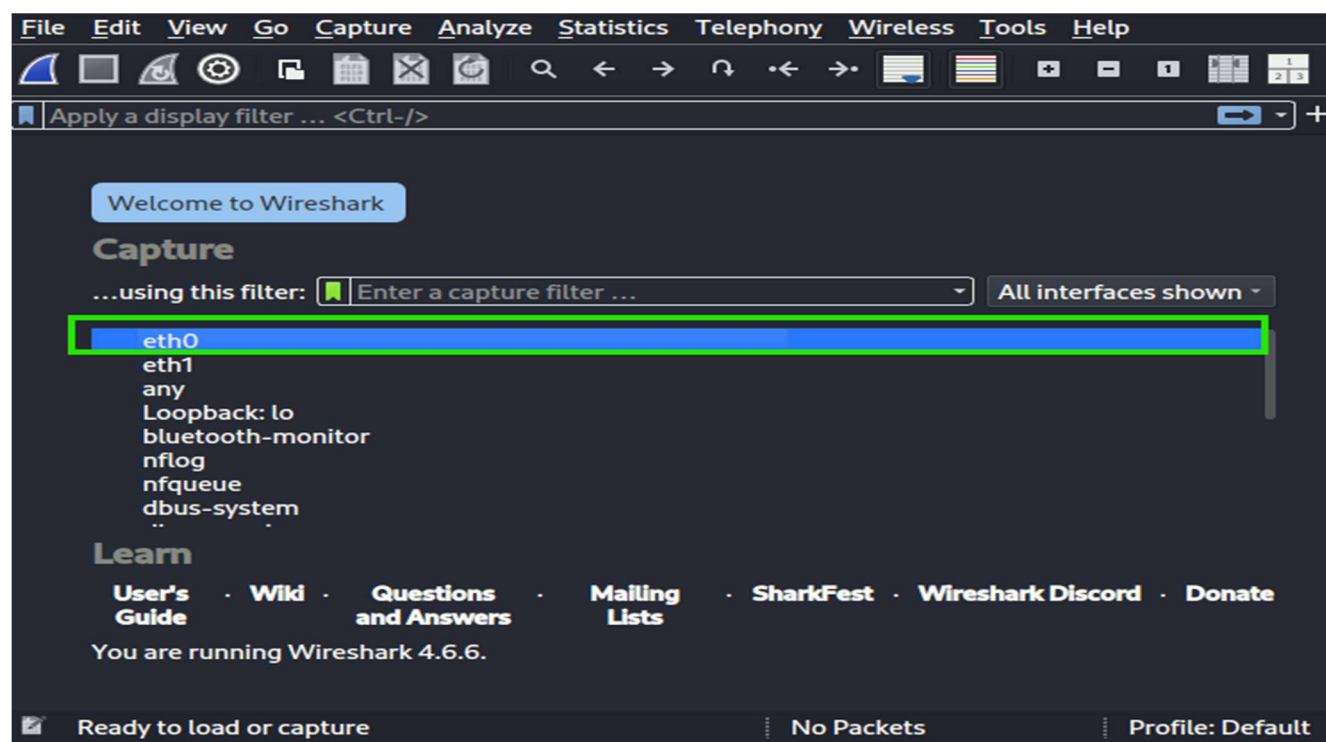
Ethernet adapter Ethernet2:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::4332:b1ff:503b:de8%25
    IPv4 Address. . . . . : 192.168.233.153
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

PS C:\Users\Kiran>

```

Step-2: Demonstrate SSH Login: Run `ssh user@server` and capture TCP/22 encrypted stream.



```
(kali@kali)-[~]
$ ssh kiran@192.168.233.149
kiran@192.168.233.149's password:
Permission denied, please try again.
kiran@192.168.233.149's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-14-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

378 updates can be applied immediately.
303 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Aug 3 18:55:02 2026 from 192.168.233.153
kiran@kiran-VMware:~$
```

```
kiran@kiran-VMware:~$ hostname
kiran-VMware
kiran@kiran-VMware:~$ pwd
/home/kiran
kiran@kiran-VMware:~$ ls
Desktop Documents Downloads Music Pictures Public snap Templates Videos
kiran@kiran-VMware:~$
```

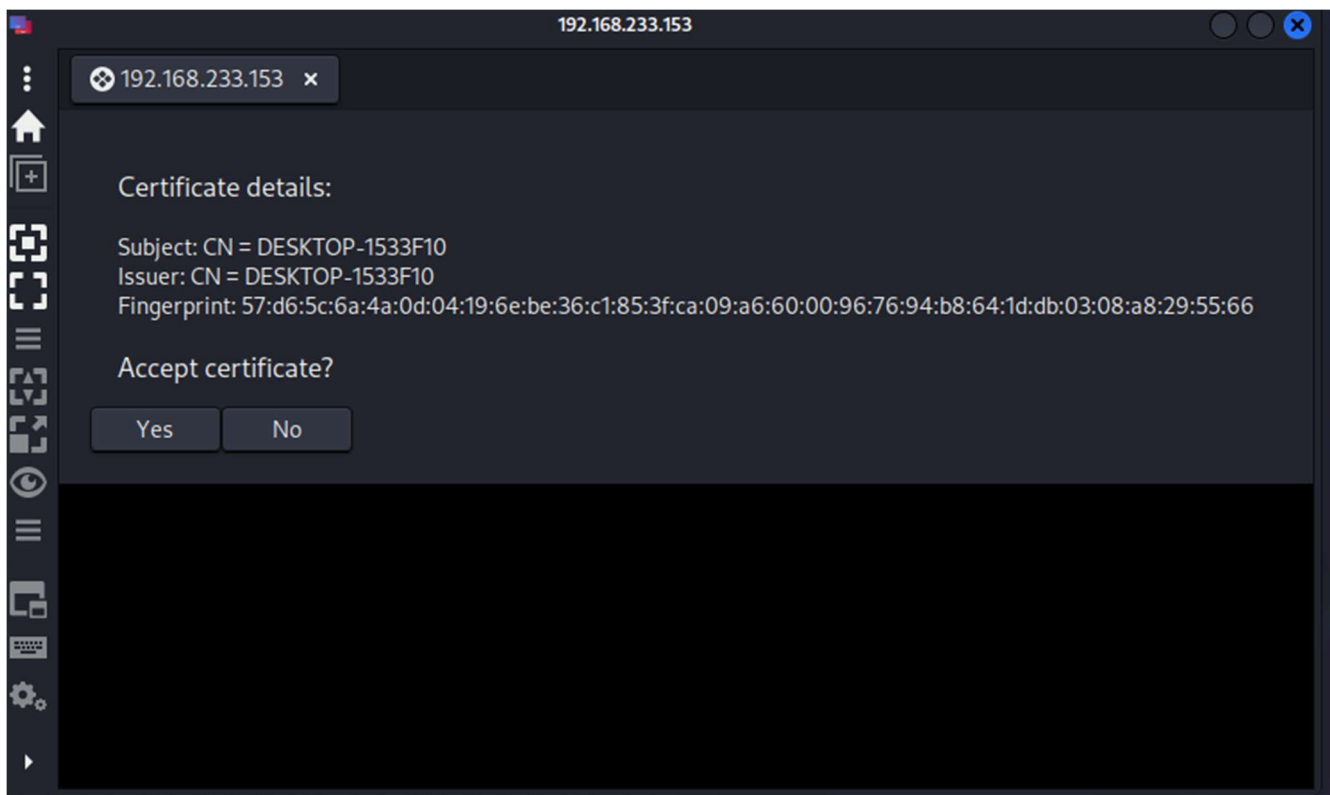
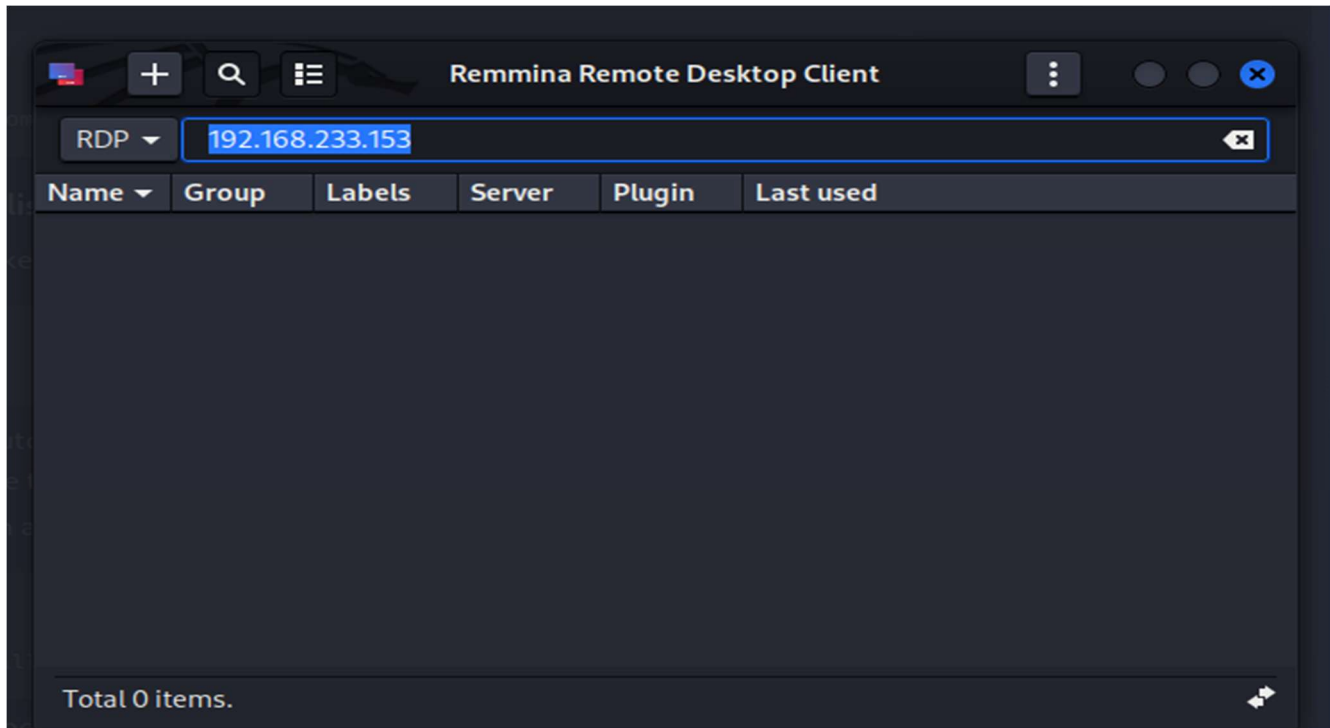
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.233.142	192.168.233.149	SSH	150	Client: Encrypted packet (len=84)
2	0.006378801	192.168.233.149	192.168.233.142	SSH	94	Server: Encrypted packet (len=28)
3	0.006399976	192.168.233.142	192.168.233.149	TCP	66	40968 → 22 [ACK] Seq=85 Ack=29 Win=252 Len=0 TSval=3453463401 TSecr=1670838551
4	0.006556875	192.168.233.142	192.168.233.149	SSH	178	Client: Encrypted packet (len=112)
5	0.048635408	192.168.233.149	192.168.233.142	TCP	66	22 → 40968 [ACK] Seq=29 Ack=197 Win=123 Len=0 TSval=1670838594 TSecr=3453463401
6	0.067948927	192.168.233.149	192.168.233.142	SSH	694	Server: Encrypted packet (len=628)
7	0.10738167	192.168.233.142	192.168.233.149	TCP	66	40968 → 22 [ACK] Seq=197 Ack=657 Win=250 Len=0 TSval=3453463506 TSecr=1670838613
8	0.110945424	192.168.233.149	192.168.233.142	SSH	110	Server: Encrypted packet (len=44)
9	0.11096823	192.168.233.142	192.168.233.149	TCP	66	40968 → 22 [ACK] Seq=197 Ack=701 Win=250 Len=0 TSval=3453463506 TSecr=1670838656
10	0.111043446	192.168.233.142	192.168.233.149	SSH	594	Client: Encrypted packet (len=528)
11	0.111169377	192.168.233.149	192.168.233.142	TCP	66	22 → 40968 [ACK] Seq=701 Ack=725 Win=122 Len=0 TSval=1670838656 TSecr=3453463506
12	0.112317009	192.168.233.149	192.168.233.142	SSH	174	Server: Encrypted packet (len=108)
13	0.113004196	192.168.233.149	192.168.233.142	SSH	718	Server: Encrypted packet (len=652)
14	0.113047716	192.168.233.142	192.168.233.149	TCP	66	40968 → 22 [ACK] Seq=725 Ack=1461 Win=250 Len=0 TSval=3453463508 TSecr=1670838657
15	0.130312368	192.168.233.149	192.168.233.142	SSH	182	Server: Encrypted packet (len=116)
16	0.170716496	192.168.233.142	192.168.233.149	TCP	66	40968 → 22 [ACK] Seq=725 Ack=1577 Win=250 Len=0 TSval=3453463566 TSecr=1670838675

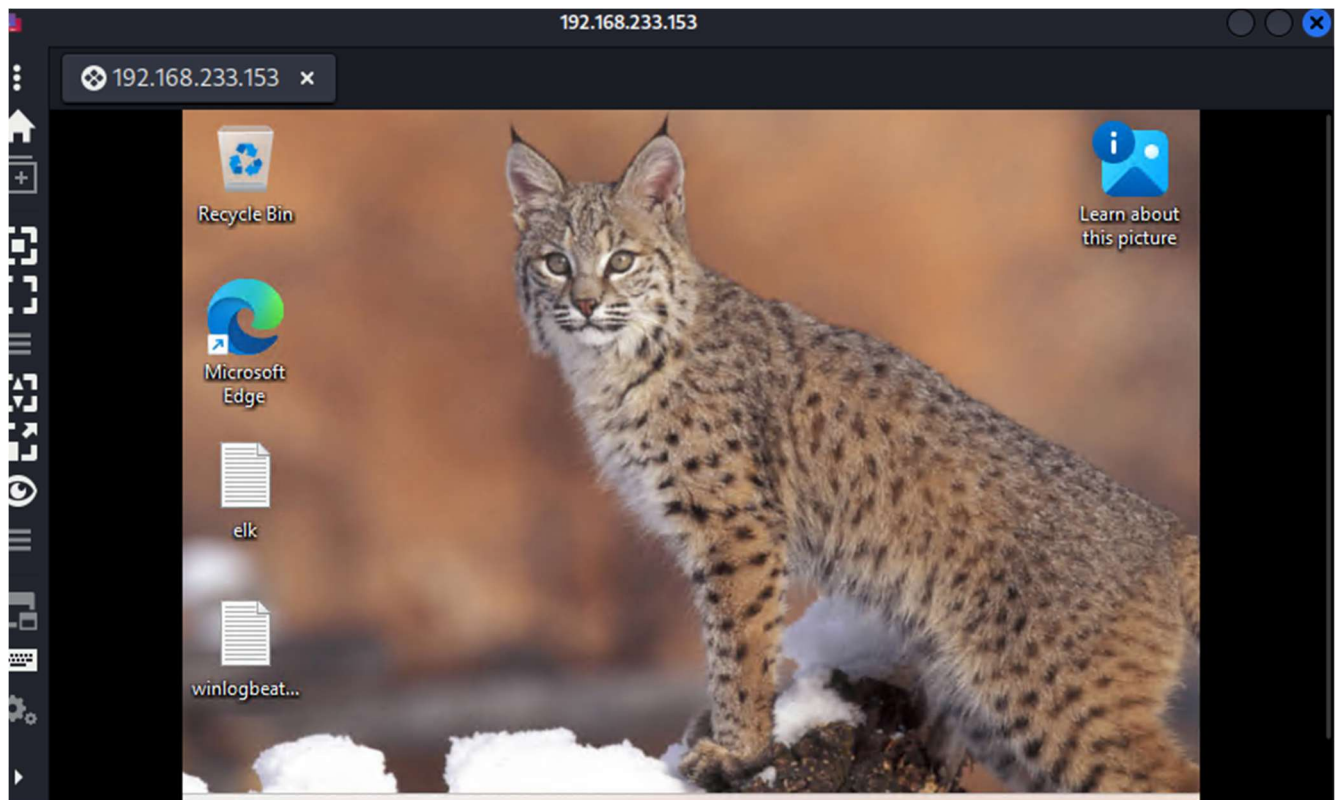
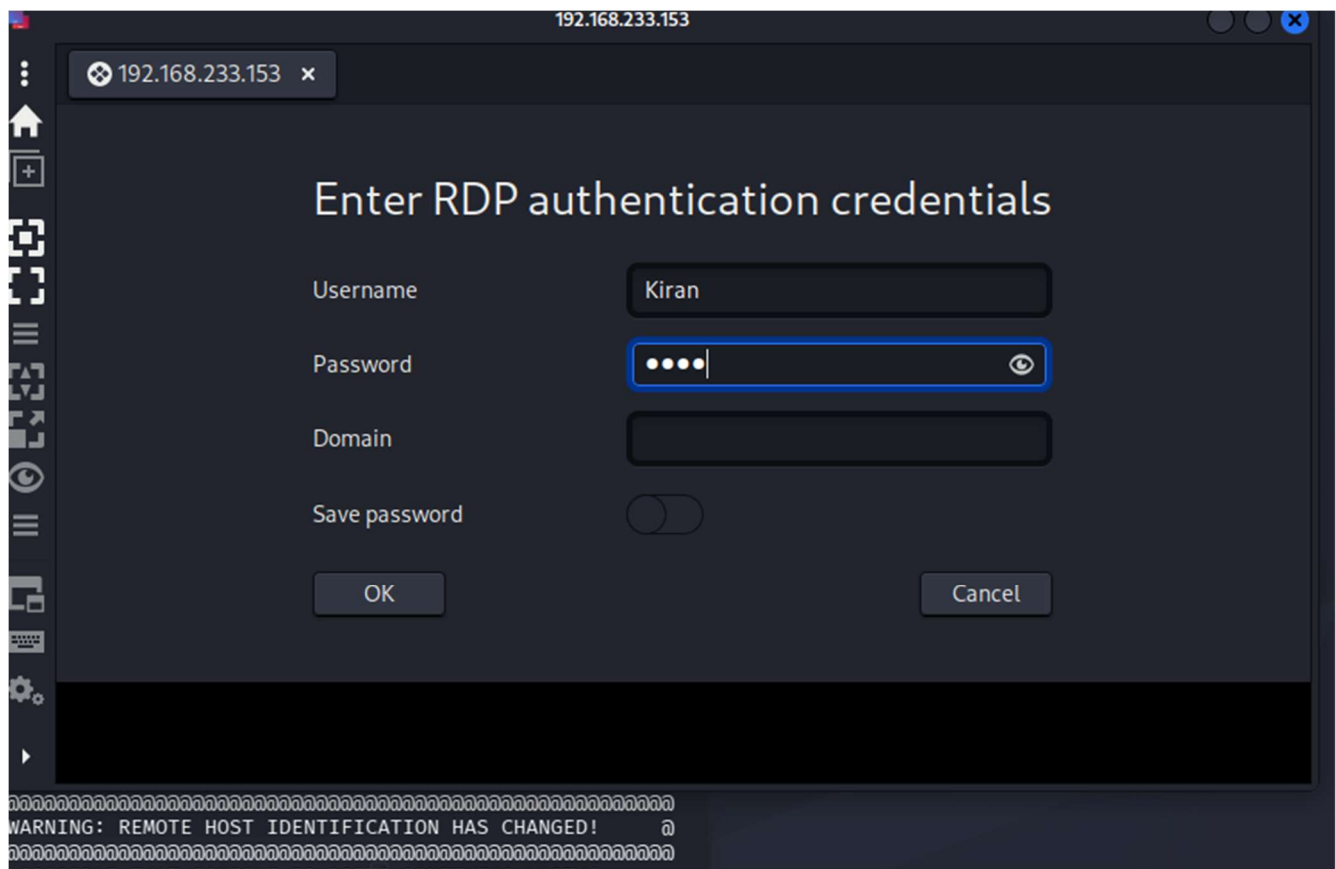
Save these wireshark capture as : SSH_Login.pcapng

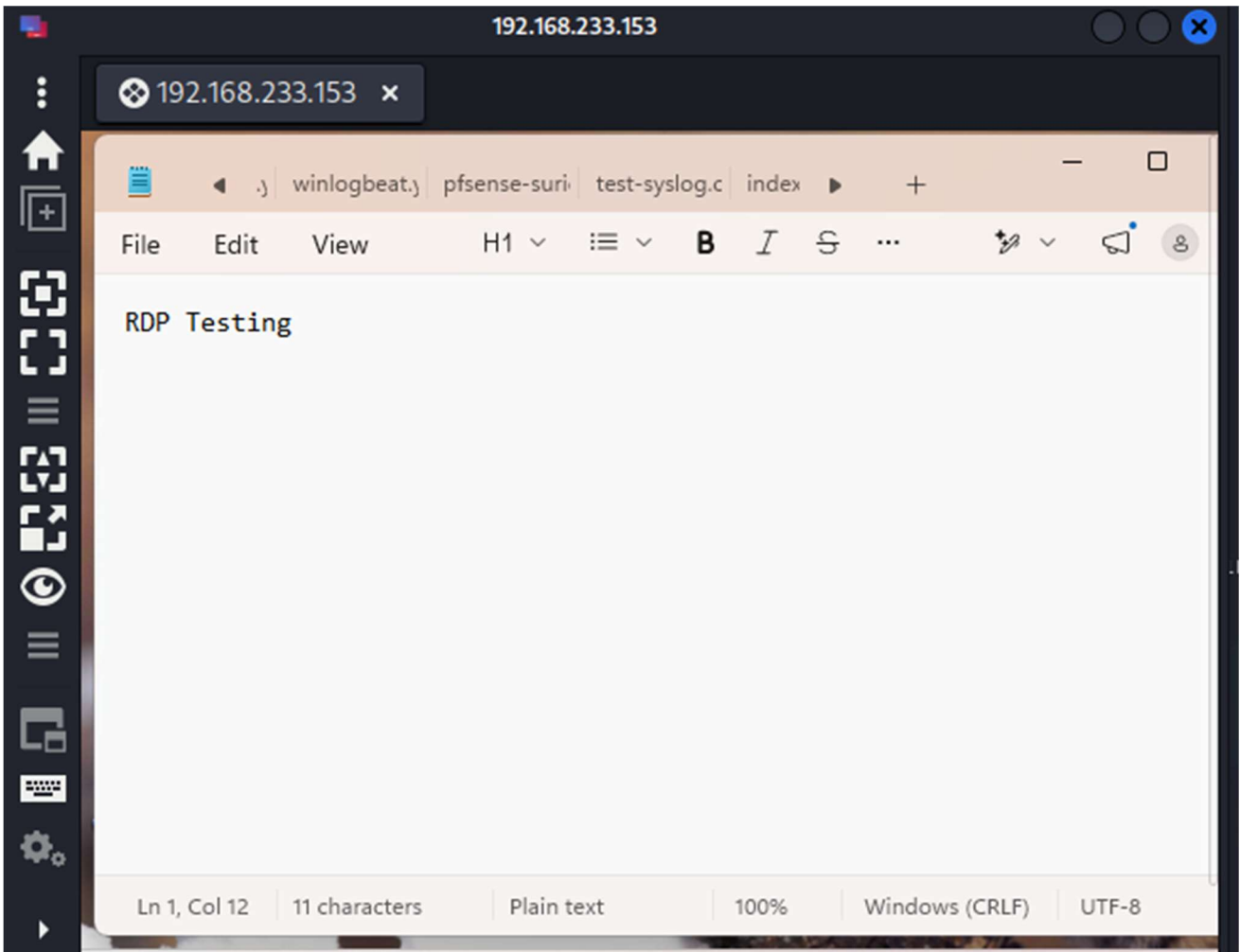
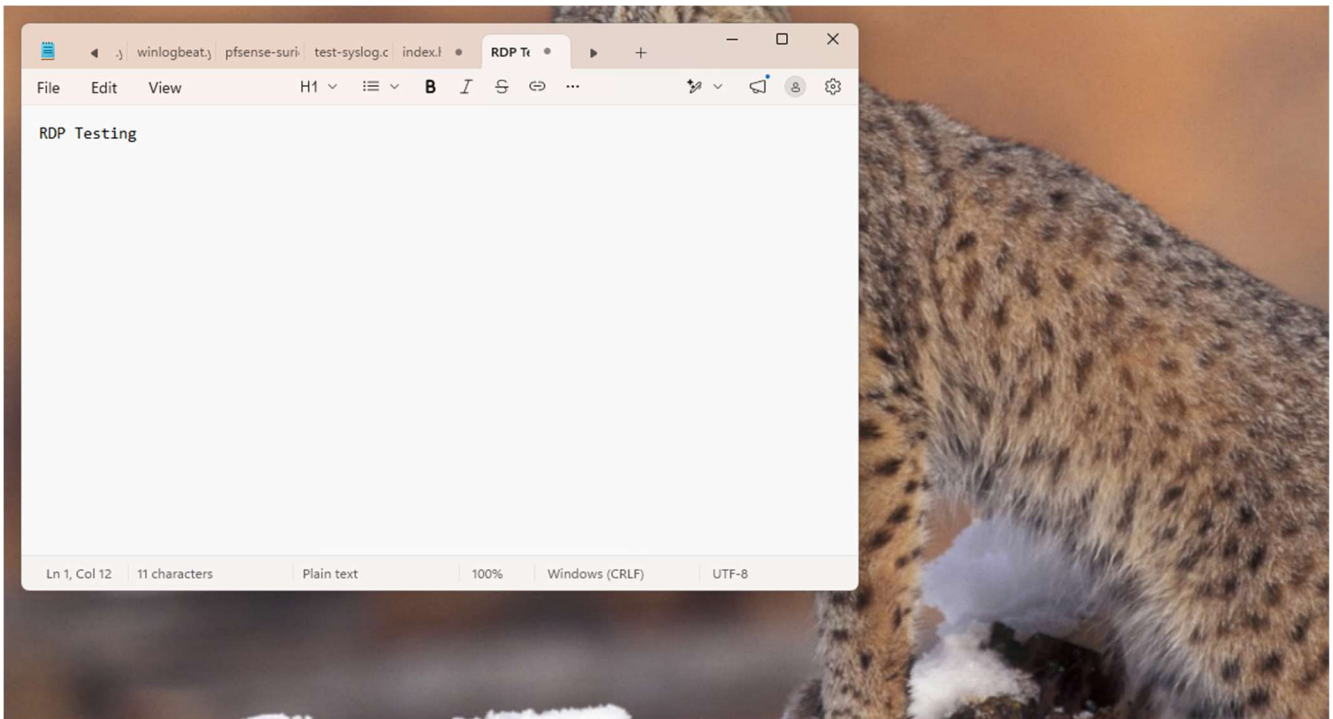
Step-3: Demonstrate RDP Login: Connect via `mstsc` and capture TCP/3389 TLS/NLA stream.

```
(kali@kali)-[~]
$ remmina
remmina
Command 'remmina' not found, but can be installed with:
sudo apt install remmina
Do you want to install it? (N/y)y
sudo apt install remmina
[sudo] password for kali:
The following packages were automatically installed and are no longer required:
bluez-hcidump libavc1394-0 libipc-run-perl libpoppler147 libsd2-classic libteamdctl0 libxmlsec1-1 openjdk-21-jre python3-pluginbase
enchant-2 libgdal38 libimbedcrypto16 libpostal1 libsimdutf31 libunibreak6 libxmlsec1-openssl1 openjdk-21-jre-headless python3-pysnmp4
firmware-marvell-presteria libio-pty-perl libntfs-3g89t64 libraw23t64 libsodium23 libvidstab1.1 network-manager-gnome postgresql-common-dev python3-tz
Use 'sudo apt autoremove' to remove them.
Installing:
remmina
```

```
(kali@kali)~$ remmina
Please share the output of these commands:
remmina-Message: 10:28:55.185: Remmina does not log all output statements. Turn on more verbose output by using "G_MESSAGES_DEBUG=remmina" as an environment variable.
More info available on the Remmina wiki at:
https://gitlab.com/Remmina/Remmina/-/wikis/Usage/Remmina-debugging
(process:123217): Gdk-CRITICAL **: 10:28:55.187: gdk_display_get_primary_monitor: assertion 'GDK_IS_DISPLAY (display)' failed
(process:123217): Gdk-CRITICAL **: 10:28:55.187: gdk_display_get_n_monitors: assertion 'GDK_IS_DISPLAY (display)' failed
```







Step-4: Demonstrate VPN & VDI Logins: Establish OpenVPN tunnel, then launch web VDI session; save and review Wireshark traces.

```
kiran@kiran-VMware:~$ sudo apt install openvpn easy-rsa -y
[sudo] password for kiran:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libccid libeac3 opensc opensc-pkcs11 pcscd
Suggested packages:
  openvpn-dco-dkms openvpn-systemd-resolved
The following NEW packages will be installed:
  easy-rsa libccid libeac3 opensc opensc-pkcs11 pcscd
The following packages will be upgraded:
  openvpn
1 upgraded, 6 newly installed, 0 to remove and 369 not upgraded.
Need to get 2.257 kB of archives.
```

```
kiran@kiran-VMware:~$ make-cadir ~/openvpn-ca
kiran@kiran-VMware:~$ cd ~/openvpn-ca
kiran@kiran-VMware:~/openvpn-ca$ ./easyrsa init-pki
```

Notice

'init-pki' complete; you may now create a CA or requests.

Your newly created PKI dir is:

* /home/kiran/openvpn-ca/pki

Using Easy-RSA configuration:

* /home/kiran/openvpn-ca/vars

Notice

DH parameters of size 2048 created at:

* /home/kiran/openvpn-ca/pki/dh.pem

```
kiran@kiran-VMware:~/openvpn-ca$ openvpn --genkey secret ta.key
kiran@kiran-VMware:~/openvpn-ca$ sudo cp pki/ca.crt /etc/openvpn
kiran@kiran-VMware:~/openvpn-ca$ sudo cp pki/private/server.key /etc/openvpn
kiran@kiran-VMware:~/openvpn-ca$ sudo cp pki/issued/server.crt /etc/openvpn
kiran@kiran-VMware:~/openvpn-ca$ sudo cp pki/dh.pem /etc/openvpn
kiran@kiran-VMware:~/openvpn-ca$ sudo cp ta.key /etc/openvpn
kiran@kiran-VMware:~/openvpn-ca$
```

```
kiran@kiran-VMware:~/openvpn-ca$ sudo nano /etc/openvpn/server.conf
```

```
GNU nano 7.2 /etc/openvpn/server.conf *
port 1194

proto udp

dev tun

ca ca.crt

cert server.crt

key server.key

dh dh.pem

server 10.8.0.0 255.255.255.0

push "redirect-gateway def1"

push "dhcp-option DNS 8.8.8.8"

keepalive 10 120

tls-auth ta.key 0

cipher AES-256-CBC
```

```
cipher AES-256-CBC
```

```
persist-key
```

```
persist-tun
```

```
verb 3
```

```
kiran@kiran-VMware:~/openvpn-ca$ sudo nano /etc/sysctl.conf
```

```
# Uncomment the following to stop low-level messages on console
#kernel.printk = 3 4 1 3

#####
# Functions previously found in netbase
#

# Uncomment the next two lines to enable Spoof protection (reverse-path filter)
# Turn on Source Address Verification in all interfaces to
# prevent some spoofing attacks
#net.ipv4.conf.default.rp_filter=1
#net.ipv4.conf.all.rp_filter=1

# Uncomment the next line to enable TCP/IP SYN cookies
# See http://lwn.net/Articles/277146/
# Note: This may impact IPv6 TCP sessions too
#net.ipv4.tcp_syncookies=1

# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=1

# Uncomment the next line to enable packet forwarding for IPv6
# Enabling this option disables Stateless Address Autoconfiguration
# based on Router Advertisements for this host
#net.ipv6.conf.all.forwarding=1
```

```
kiran@kiran-VMware:~/openvpn-ca$ sudo sysctl -p
net.ipv4.ip_forward = 1
kiran@kiran-VMware:~/openvpn-ca$
```

```
kiran@kiran-VMware:~/openvpn-ca$ sudo systemctl start openvpn@server
kiran@kiran-VMware:~/openvpn-ca$ sudo systemctl enable openvpn@server
Created symlink /etc/systemd/system/multi-user.target.wants/openvpn@server.service → /usr/lib/systemd/system/openvpn@server.service.
kiran@kiran-VMware:~/openvpn-ca$ sudo systemctl status openvpn@server
● openvpn@server.service - OpenVPN connection to server
   Loaded: loaded (/usr/lib/systemd/system/openvpn@server.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-08-04 19:51:43 IST; 12s ago
     Docs: man:openvpn(8)
           https://community.openvpn.net/openvpn/wiki/Openvpn24ManPage
           https://community.openvpn.net/openvpn/wiki/HOWTO
   Main PID: 26666 (openvpn)
    Status: "Initialization Sequence Completed"
     Tasks: 1 (limit: 10)
    Memory: 1.4M (peak: 1.7M)
       CPU: 15ms
    CGroup: /system.slice/system-openvpn.slice/openvpn@server.service
            └─26666 /usr/sbin/openvpn --daemon ovpn-server --status /run/openvpn/server.status 10 --cd /etc/openvpn --
```

Aug 04 19:51:43 kiran-VMware ovpn-server[26666]: net_iface_up: set tun0 up


```
GNU nano 7.2                                kali.ovpn *
client
dev tun
proto udp
remote 192.168.233.132 1194
resolv-retry infinite
nobind
persist-key
persist-tun
ca ca.crt
cert kali.crt
key kali.key
tls-auth ta.key 1
cipher AES-256-CBC
```

```
cipher AES-256-CBC
```

```
verb 3
```

```
(kali@kali)-[~]
$ scp kiran@192.168.233.132:/home/kiran/openvpn-ca/kali.ovpn .
kiran@192.168.233.132's password:
kali.ovpn 100% 200 248.2KB/s 00:00
(kali@kali)-[~]
```

```
(kali@kali)-[~]
$ scp kiran@192.168.233.132:/home/kiran/openvpn-ca/pki/ca.crt .
kiran@192.168.233.132's password:
ca.crt 100% 1180 1.7MB/s 00:00
(kali@kali)-[~]
```

```
(kali@kali)-[~]
$ scp kiran@192.168.233.132:/home/kiran/openvpn-ca/ta.key .
kiran@192.168.233.132's password:
ta.key 100% 636 864.3KB/s 00:00
(kali@kali)-[~]
```

```
(kali@kali)-[~]
$ scp kiran@192.168.233.132:/home/kiran/openvpn-ca/pki/issued/kali.crt .
kiran@192.168.233.132's password:
kali.crt
```

```
(kali㉿kali)-[~]
$ scp kiran@192.168.233.132:/home/kiran/openvpn-ca/pki/private/kali.key .
kiran@192.168.233.132's password:
kali.key
```

```
(kali㉿kali)-[~]
$ sudo apt update
[sudo] password for kali:
Hit:1 http://http.kali.org/kali kali-rolling InRelease
63 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali㉿kali)-[~]
$ sudo apt install openvpn -y
openvpn is already the newest version (2.7.5-1).
openvpn set to manually installed.
The following packages were automatically installed and are no longer required:
bluez-hcidump libavc1394-0 libbpc-run-perl libbpoopler147 libbsd12-classic libteamdctl0 libxmlsec1-1 openjdk-21-jre python3-pluginbase
enchant-2 libgdal38 libbmedcryptol6 libpostall1 libsimdutf31 libunibreak6 libxmlsec1-openssl1 openjdk-21-jre-headless python3-pysnmp4
firmware-marvell-prestera libio-pty-perl libntfs-3g89t64 libraw23t64 libsodium23 libvidstab1.1 network-manager-gnome postgresql-common-dev python3-tz
Use 'sudo apt autoremove' to remove them.

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 63

(kali㉿kali)-[~]
$
```

```
(kali㉿kali)-[~/Downloads]
$ sudo openvpn kali.ovpn
2026-08-04 10:52:01 DEPRECATED OPTION: --persist-key option ignored. Keys are now always persisted across restarts.
2026-08-04 10:52:01 DEPRECATED OPTION: --cipher set to 'AES-256-CBC' but missing in --data-ciphers (DEFAULT). OpenVPN ignores --cipher for cipher negotiations.
2026-08-04 10:52:01 OpenVPN 2.7.5 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AEAD] [DCO]
2026-08-04 10:52:01 Library versions: OpenSSL 3.6.3 9 Jun 2026, LZO 2.10
2026-08-04 10:52:01 DCO version: 7.0.12-kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 7.0.12-2kali1 (2026-06-18)
2026-08-04 10:52:01 WARNING: No server certificate verification method has been enabled. See http://openvpn.net/howto.html#mtm for more info.
2026-08-04 10:52:01 TCP/UDP: Preserving recently used remote address: [AF_INET]192.168.233.132:1194
2026-08-04 10:52:01 Socket Buffers: R=[212992→212992] S=[212992→212992]
2026-08-04 10:52:01 UDPv4 link local: (not bound)
2026-08-04 10:52:01 UDPv4 link remote: [AF_INET]192.168.233.132:1194
2026-08-04 10:52:01 TLS: Initial packet from [AF_INET]192.168.233.132:1194, sid=0d9c64d1 916edd49
2026-08-04 10:52:01 VERIFY OK: depth=1, CN=kiran
2026-08-04 10:52:01 VERIFY OK: depth=0, CN=kiran
2026-08-04 10:52:01 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384, peer certificate: 2048 bits RSA, signature: RSA-SHA256, peer temporary key: 253 bits
2026-08-04 10:52:01 [kiran] Peer Connection Initiated with [AF_INET]192.168.233.132:1194
2026-08-04 10:52:01 TLS: move_session: dest=TM_ACTIVE src=TM_INITIAL reinit_src=1
2026-08-04 10:52:01 TLS: tls_multi process: initial untrusted session promoted to trusted
2026-08-04 10:52:01 PUSH: Received control message: 'PUSH_REPLY,redirect-gateway def1,dhcp-option DNS 8.8.8.8,route 10.8.0.1,topology net30,ping 10,ping-restart 120,ifconfig-s256-GCM,protocol-flags cc-exit tls-ekm dyn-tls-crypt,tun-mtu 1500'
2026-08-04 10:52:01 OPTIONS IMPORT: --ifconfig/up options modified
2026-08-04 10:52:01 OPTIONS IMPORT: route options modified
2026-08-04 10:52:01 OPTIONS IMPORT: --ip-win32 and/or --dhcp-option options modified
2026-08-04 10:52:01 OPTIONS IMPORT: tun-mtu set to 1500
2026-08-04 10:52:01 net_route_v4_best_gw query: dst 0.0.0.0
2026-08-04 10:52:01 net_route_v4_best_gw result: via 192.168.22.2 dev eth0
2026-08-04 10:52:01 net_route_v4_best_gw query: dst 192.168.233.132
2026-08-04 10:52:01 net_route_v4_best_gw result: via 0.0.0.0 dev eth1
2026-08-04 10:52:01 net_iface_new: add tun0 type ovpn
2026-08-04 10:52:01 DCO device tun0 opened
2026-08-04 10:52:01 ovpn-dco device [tun0] opened
2026-08-04 10:52:01 net_iface_mtu_set: mtu 1500 for tun0
2026-08-04 10:52:01 net_iface_up: set tun0 up
2026-08-04 10:52:01 net_addr_ptp_v4_add: 10.8.0.6 peer 10.8.0.5 dev tun0
2026-08-04 10:52:01 /usr/libexec/openvpn/dns-updown
setting DNS using resolv.conf file
2026-08-04 10:52:01 dns_up command exited with status 0
2026-08-04 10:52:01 net_route_v4_add: 192.168.233.132/32 via 0.0.0.0 dev eth1 table 0 metric -1
2026-08-04 10:52:01 net_route_v4_add: 0.0.0.0/1 via 10.8.0.5 dev [NULL] table 0 metric -1
2026-08-04 10:52:01 net_route_v4_add: 128.0.0.0/1 via 10.8.0.5 dev [NULL] table 0 metric -1
2026-08-04 10:52:01 net_route_v4_add: 10.8.0.1/32 via 10.8.0.5 dev [NULL] table 0 metric 200
2026-08-04 10:52:01 Initialization Sequence Completed
2026-08-04 10:52:01 Data Channel: cipher 'AES-256-GCM', peer-id: 0
2026-08-04 10:52:01 Timers: ping 10, ping-restart 120
2026-08-04 10:52:01 Protocol options: protocol-flags cc-exit tls-ekm dyn-tls-crypt
```

```

(kali㉿kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5b:e0:bd brd ff:ff:ff:ff:ff:ff
    inet 192.168.22.133/24 brd 192.168.22.255 scope global dynamic noprefixroute eth0
        valid_lft 942sec preferred_lft 942sec
    inet6 fe80::ba50:6c33:cff6:cc7d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5b:e0:c7 brd ff:ff:ff:ff:ff:ff
    inet 192.168.233.142/24 brd 192.168.233.255 scope global dynamic noprefixroute eth1
        valid_lft 942sec preferred_lft 942sec
    inet6 fe80::f88:f516:a369:249e/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
6: tun0: <POINTOPOINT,NOARP,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/none
    inet 10.8.0.6 peer 10.8.0.5/32 scope global tun0
        valid_lft forever preferred_lft forever
    inet6 fe80::9c34:a683:775f:793e/64 scope link stable-privacy proto kernel_ll
        valid_lft forever preferred_lft forever

(kali㉿kali)-[~]

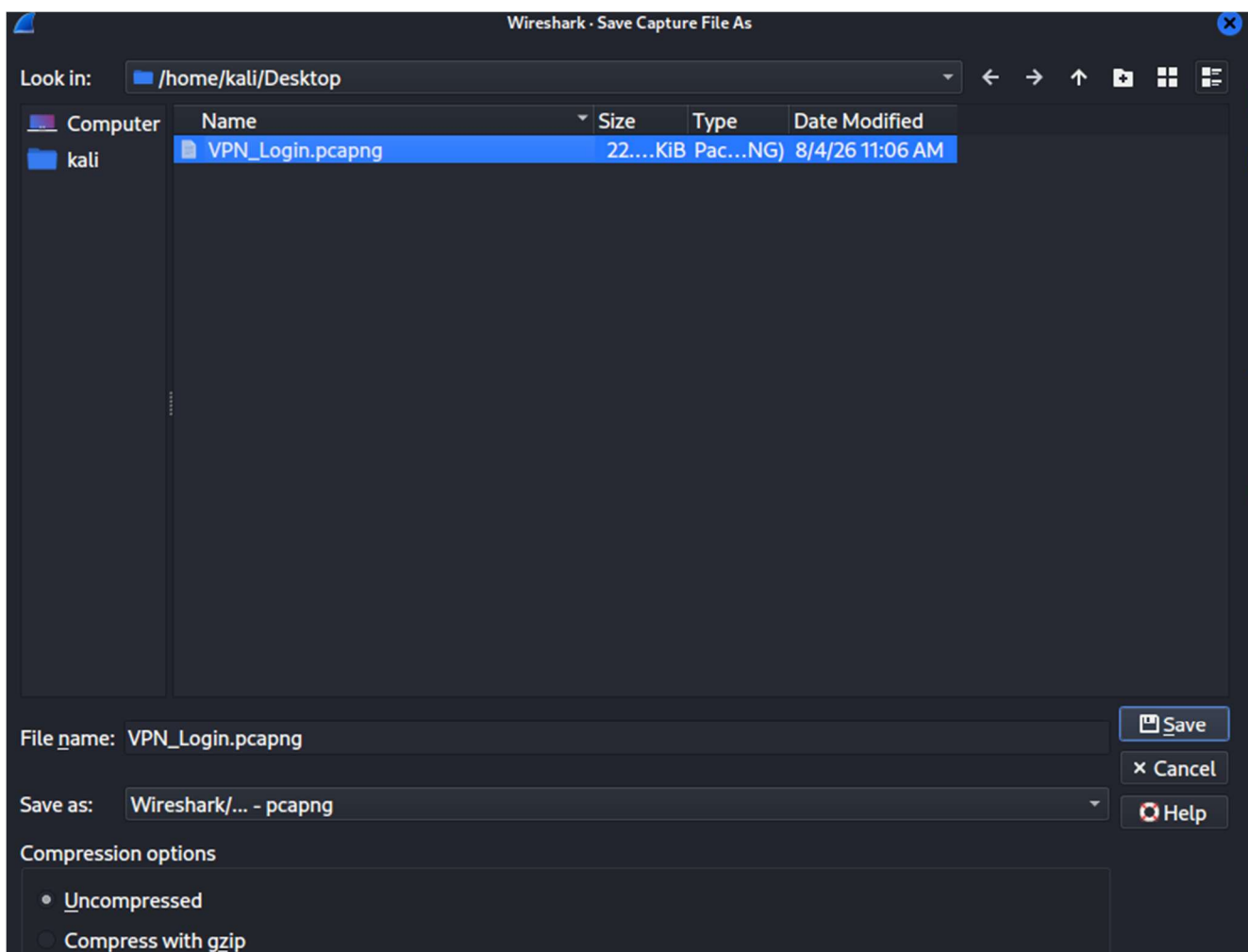
```

```

(kali㉿kali)-[~]
$ ping 10.8.0.6
PING 10.8.0.6 (10.8.0.6) 56(84) bytes of data.
64 bytes from 10.8.0.6: icmp_seq=1 ttl=64 time=0.910 ms
64 bytes from 10.8.0.6: icmp_seq=2 ttl=64 time=0.016 ms
64 bytes from 10.8.0.6: icmp_seq=3 ttl=64 time=0.023 ms
64 bytes from 10.8.0.6: icmp_seq=4 ttl=64 time=0.065 ms
64 bytes from 10.8.0.6: icmp_seq=5 ttl=64 time=0.021 ms

```

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.233.142	192.168.233.132	OpenVPN	106	MessageType: P_DATA_V2
2	0.022212601	192.168.233.142	192.168.233.132	OpenVPN	106	MessageType: P_DATA_V2
3	2.004402515	192.168.233.132	192.168.233.142	OpenVPN	82	MessageType: P_DATA_V2
4	2.004426029	192.168.233.142	192.168.233.132	ICMP	116	Destination unreachable (Port unreachable)
5	2.427263279	192.168.233.142	192.168.233.132	OpenVPN	84	MessageType: P_CONTROL_HARD_RESET_CLIENT_V2
6	2.427539315	192.168.233.132	192.168.233.142	OpenVPN	96	MessageType: P_CONTROL_HARD_RESET_SERVER_V2
7	2.427643337	192.168.233.142	192.168.233.132	OpenVPN	1264	MessageType: P_CONTROL_V1 [OpenVPN Message segment of a reassembled PDU]
8	2.427690955	192.168.233.142	192.168.233.132	TLSv1.3	443	Client Hello
9	2.428001158	192.168.233.132	192.168.233.142	OpenVPN	92	MessageType: P_ACK_V1
10	2.428734163	192.168.233.132	192.168.233.142	TLSv1.3	1264	Server Hello, Change Cipher Spec, Application Data, Application Data [OpenVPN Message segment of a reassembled PDU]
11	2.428829335	192.168.233.132	192.168.233.142	TLSv1.3	1264	Application Data, Application Data [OpenVPN Message segment of a reassembled PDU]
12	2.428829365	192.168.233.132	192.168.233.142	TLSv1.3	116	Application Data
13	2.428891766	192.168.233.142	192.168.233.132	OpenVPN	96	MessageType: P_ACK_V1
14	2.429142513	192.168.233.142	192.168.233.132	OpenVPN	100	MessageType: P_ACK_V1
15	2.429816590	192.168.233.142	192.168.233.132	TLSv1.3	1264	Change Cipher Spec [OpenVPN Message segment of a reassembled PDU]
16	2.429857598	192.168.233.142	192.168.233.132	TLSv1.3	1264	Application Data, Application Data, Application Data [OpenVPN Message segment of a reassembled PDU]
17	2.429888965	192.168.233.142	192.168.233.132	TLSv1.3	268	Application Data
18	2.430022928	192.168.233.132	192.168.233.142	OpenVPN	100	MessageType: P_ACK_V1
19	2.430417194	192.168.233.132	192.168.233.142	TLSv1.3	266	Application Data, Application Data
20	2.430473358	192.168.233.142	192.168.233.132	OpenVPN	100	MessageType: P_ACK_V1



```
kiran@kiran-VMware:~$ sudo apt install default-jdk -y
[sudo] password for kiran:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  ca-certificates-java default-jdk-headless default-jre default-jre-headless
  fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-jni
  libice-dev libpthread-stubs0-dev libsm-dev libx11-dev libxau-dev libxcb1-dev
  libxdmcp-dev libxt-dev openjdk-21-jdk openjdk-21-jdk-headless openjdk-21-jre
  openjdk-21-jre-headless x11proto-dev xorg-sgml-doctools xtrans-dev
Suggested packages:
  libice-doc libsm-doc libx11-doc libxcb-doc libxt-doc openjdk-21-demo
  openjdk-21-source visualvm fonts-ipafont-gothic fonts-ipafont-mincho
  fonts-wqy-microhei | fonts-wqy-zenhei fonts-indic
The following NEW packages will be installed:
  ca-certificates-java default-jdk default-jdk-headless default-jre
  default-jre-headless fonts-dejavu-extra java-common libatk-wrapper-java
  libatk-wrapper-java-jni libice-dev libpthread-stubs0-dev libsm-dev
  libx11-dev libxau-dev libxcb1-dev libxdmcp-dev libxt-dev openjdk-21-jdk
  openjdk-21-jdk-headless openjdk-21-jre openjdk-21-jre-headless x11proto-dev
```

```
kiran@kiran-VMware:~$ sudo apt install tomcat10 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libapache-pom-java libapr1t64 libbcel-java libcommons-compress-java
  libcommons-io-java libcommons-parent-java libeclipse-jdt-core-java
  libtcnative-2 libtomcat10-java tomcat-jakartaee-migration tomcat10-common
Suggested packages:
  libbcel-java-doc libasm-java libxz-java libcommons-io-java-doc
  tomcat10-admin tomcat10-docs tomcat10-examples tomcat10-user
The following NEW packages will be installed:
  libapache-pom-java libapr1t64 libbcel-java libcommons-compress-java
  libcommons-io-java libcommons-parent-java libeclipse-jdt-core-java
  libtcnative-2 libtomcat10-java tomcat-jakartaee-migration tomcat10
```

```

kiran@kiran-VMware:~$ sudo apt install docker.io docker-compose -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bridge-utils containerd git git-man liberror-perl pigz python3-compose
  python3-docker python3-dockerpty python3-docopt python3-dotenv
  python3-packaging python3-texttable python3-websocket runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools btrfs-progs cgroupfs-mount | cgroup-lite debootstrap
  docker-buildx docker-compose-v2 docker-doc rinse zfs-fuse | zfsutils
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb
  git-cvs git-mediawiki git-svn
The following NEW packages will be installed:
  bridge-utils containerd docker-compose docker.io git git-man liberror-perl
  pigz python3-compose python3-docker python3-dockerpty python3-docopt
  python3-dotenv python3-packaging python3-texttable python3-websocket runc
  ubuntu-fan
0 upgraded, 18 newly installed, 0 to remove and 2 not upgraded.
Need to get 78.6 MB of archives.

```

```

kiran@kiran-VMware:~$ sudo apt install tomcat10 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libapache-pom-java libapr1t64 libbcel-java libcommons-compress-java
  libcommons-io-java libcommons-parent-java libeclipse-jdt-core-java
  libtcnative-2 libtomcat10-java tomcat-jakartaee-migration tomcat10-common
Suggested packages:
  libbcel-java-doc libasm-java libxz-java libcommons-io-java-doc
  tomcat10-admin tomcat10-docs tomcat10-examples tomcat10-user
The following NEW packages will be installed:
  libapache-pom-java libapr1t64 libbcel-java libcommons-compress-java
  libcommons-io-java libcommons-parent-java libeclipse-jdt-core-java
  libtcnative-2 libtomcat10-java tomcat-jakartaee-migration tomcat10
  tomcat10-common
0 upgraded, 12 newly installed, 0 to remove and 2 not upgraded.
Need to get 15.2 MB of archives.
After this operation, 18.1 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 libapache-pom-java all 29-2 [5,284 B]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 libcommons-parent-java all 56-1 [10.7 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 libbcel-java all 6.5.0-2 [626 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 libcommons-compress-java all 1.25.0-1 [612 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 libcommons-io-java all 2.11.0-2 [297 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/universe amd64 libeclipse-jdt-core-java all 3.32.0+eclipse4.26-2 [6,438 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 libtcnative-2 amd64 2.0.14-1~24.04.1 [48.2 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 libtomcat10-java all 10.1.55-1ubuntu2~24.04.1 [6,432 kB]
76% [9 libtomcat10-java 3,820 kB/6,432 kB 59%]
628 kB/s 5s

```

```

kiran@kiran-VMware:~$ sudo apt install guacd tomcat10 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
tomcat10 is already the newest version (10.1.55-1ubuntu2~24.04.1).
The following additional packages will be installed:
  libfreerdp-client2-2t64 libfreerdp2-2t64 libguac-client-rdp0t64 libguac-client-ssh0t64 libguac-client-vnc0t64
  libguac19t64 libossp-uuid16 libssh2-1t64 libvncclient1 libwinpr2-2t64
Suggested packages:
  libguac-client-telnet0t64 freerdp2-x11 uuid
Recommended packages:
  libfreerdp-plugins-standard vnc4server
The following NEW packages will be installed:
  guacd libfreerdp-client2-2t64 libfreerdp2-2t64 libguac-client-rdp0t64 libguac-client-ssh0t64 libguac-client-vnc0t64
  libguac19t64 libossp-uuid16 libssh2-1t64 libvncclient1 libwinpr2-2t64
0 upgraded, 11 newly installed, 0 to remove and 2 not upgraded.
Need to get 1,668 kB of archives.
After this operation, 5,510 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 libossp-uuid16 amd64 1.6.2-1.5build13 [30.9 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 libguac19t64 amd64 1.3.0-1.3ubuntu1 [37.2 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 guacd amd64 1.3.0-1.3ubuntu1 [21.3 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 libwinpr2-2t64 amd64 2.11.5+dfsg1-1build2 [350 kB]

```

```

Processing triggers for libc-bin (2.40-0ubuntu2~24.04.1) ...
kiran@kiran-VMware:~$ sudo systemctl start tomcat10
kiran@kiran-VMware:~$ sudo systemctl enable tomcat10
kiran@kiran-VMware:~$ sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Server
   Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-08-05 12:24:46 IST; 2min 41s ago
     Docs: https://tomcat.apache.org/tomcat-10.0-doc/index.html
  Main PID: 8761 (java)
    Tasks: 35 (limit: 3191)
   Memory: 234.5M (peak: 234.9M)
      CPU: 2.416s
   CGroup: /system.slice/tomcat10.service
           └─8761 /usr/lib/jvm/java-21-openjdk-amd64/bin/java -Djava.util.logging.config.file=/var/lib/tomcat10/conf/

Aug 05 12:24:46 kiran-VMware tomcat10[8761]: OpenSSL successfully initialized [OpenSSL 3.0.13 30 Jan 2024]
Aug 05 12:24:47 kiran-VMware tomcat10[8761]: Initializing ProtocolHandler ["http-nio-8080"]
Aug 05 12:24:47 kiran-VMware tomcat10[8761]: Server initialization in [691] milliseconds
Aug 05 12:24:47 kiran-VMware tomcat10[8761]: Starting service [Catalina]
Aug 05 12:24:47 kiran-VMware tomcat10[8761]: Starting Servlet engine: [Apache Tomcat/10.1.55 (Ubuntu)]
Aug 05 12:24:47 kiran-VMware tomcat10[8761]: Deploying web application directory [/var/lib/tomcat10/webapps/ROOT]
Aug 05 12:24:47 kiran-VMware tomcat10[8761]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug
Aug 05 12:24:47 kiran-VMware tomcat10[8761]: Deployment of web application directory [/var/lib/tomcat10/webapps/ROOT] h
Aug 05 12:24:47 kiran-VMware tomcat10[8761]: Starting ProtocolHandler ["http-nio-8080"]
Aug 05 12:24:47 kiran-VMware tomcat10[8761]: Server startup in [625] milliseconds
lines 1-21/21 (END)
kiran@kiran-VMware:~$

```

```

kiran@kiran-VMware:~$ sudo systemctl enable docker
kiran@kiran-VMware:~$ sudo systemctl start docker
kiran@kiran-VMware:~$ docker compose version
Docker Compose version 2.40.3+ds1-0ubuntu1~24.04.1
kiran@kiran-VMware:~$ mkdir ~/guacamole
kiran@kiran-VMware:~$ cd ~/guacamole
kiran@kiran-VMware:~/guacamole$ nano docker-compose.yml
kiran@kiran-VMware:~/guacamole$

```

```
services:
  guacd:
    image: guacamole/guacd:latest
    container_name: guacd
    restart: unless-stopped

  guacamole:
    image: guacamole/guacamole:latest
    container_name: guacamole
    restart: unless-stopped
    ports:
      - "8080:8080"
    environment:
      GUACD_HOSTNAME: guacd
    depends_on:
      - guacd
```

```
kiran@kiran-VMware:~/guacamole$ sudo docker compose up -d
```

```
[+] Running 21/21
```

```
✓ guacamole Pulled
  ✓ 888eae77723a Pull complete
  ✓ 966c395d29cb Pull complete
  ✓ d7358a7dc71c Pull complete
  ✓ cf701402aa2e Pull complete
  ✓ 28330c5e0bc9 Pull complete
  ✓ e36d95e6fd7d Pull complete
  ✓ b925c72d92d1 Pull complete
  ✓ 4f4fb700ef54 Pull complete
  ✓ fc4f1d2bc539 Pull complete
  ✓ 34d7dc22045a Pull complete
  ✓ af4cff218c88 Pull complete
  ✓ 9c6428d5b1c7 Pull complete
  ✓ 37432ea0cfc1 Pull complete
```

```
✓ guacd Pulled
  ✓ 5d3884df3c03 Pull complete
  ✓ ac9e792add43 Pull complete
  ✓ 878cd0b45d6b Pull complete
  ✓ 72c8791ea893 Pull complete
  ✓ ed089d9bb9b3 Pull complete
  ✓ 44cf07d57ee4 Pull complete
```

```
[+] Running 2/3
```

```
✓ Network guacamole_default Created
✓ Container guacd Started
```

Click or tap on a connection below to manage that connection. Depending on your access level, connections can be added and deleted, and their properties (protocol, hostname, port, etc.) can be changed.

New ConnectionNew GroupImport

Filter



EDIT CONNECTION

guacadmin

Name:Windows-VDI-Desktop

Location:ROOT

Protocol:RDP

CONCURRENCY LIMITS

Maximum number of connections:

Maximum number of connections per user:

LOAD BALANCING

Connection weight:

Use for failover only:

GUACAMOLE PROXY PARAMETERS (GUACD)

Hostname:

Port:4822

Encryption:

PARAMETERS

Network

Hostname:192.168.56.126

Port:3389

PARAMETERS

Network

Hostname:192.168.56.126

Port:3389

Connection timeout:

Authentication

Username:Kiran4

Password:*****

Domain:

Security mode:Any

Disable authentication:

Ignore server certificate:

Trust host certificate on first use:

Fingerprints of trusted host certificates:

Remote Desktop Gateway

Hostname:

Port:

Username:

Password:*****

Domain:

Wake-on-LAN (WoL)

Send WoL packet: ☐

MAC address of the remote host:

Broadcast address for WoL packet:

UDP port for WoL packet:

Host boot wait time:

Save

Clone

Cancel

Delete

USER HISTORY

EDIT USER

guacadmin

Username:

Password:

Re-enter Password:

Login disabled: ☐

PROFILE

Full name:

Email address:

Organization:

Role:

ACCOUNT RESTRICTIONS

Password expired: ☐

Allow access after:

Do not allow access after:

Enable account after:

Disable account after:

User time zone:

PERMISSIONS

Administer system: ☐

Audit system: ☐

Create new users: ☐

EDIT USER

guacadmin

Username:

vduser

Password:

Re-enter Password:

Login disabled:

☐

PROFILE

Full name:

Email address:

Organization:

Role:

ACCOUNT RESTRICTIONS

Password expired:

☐

Allow access after:

-- : -- --

Do not allow access after:

-- : -- --

Enable account after:

mm / dd / yyyy

Disable account after:

mm / dd / yyyy

User time zone:

PERMISSIONS

Administer system:

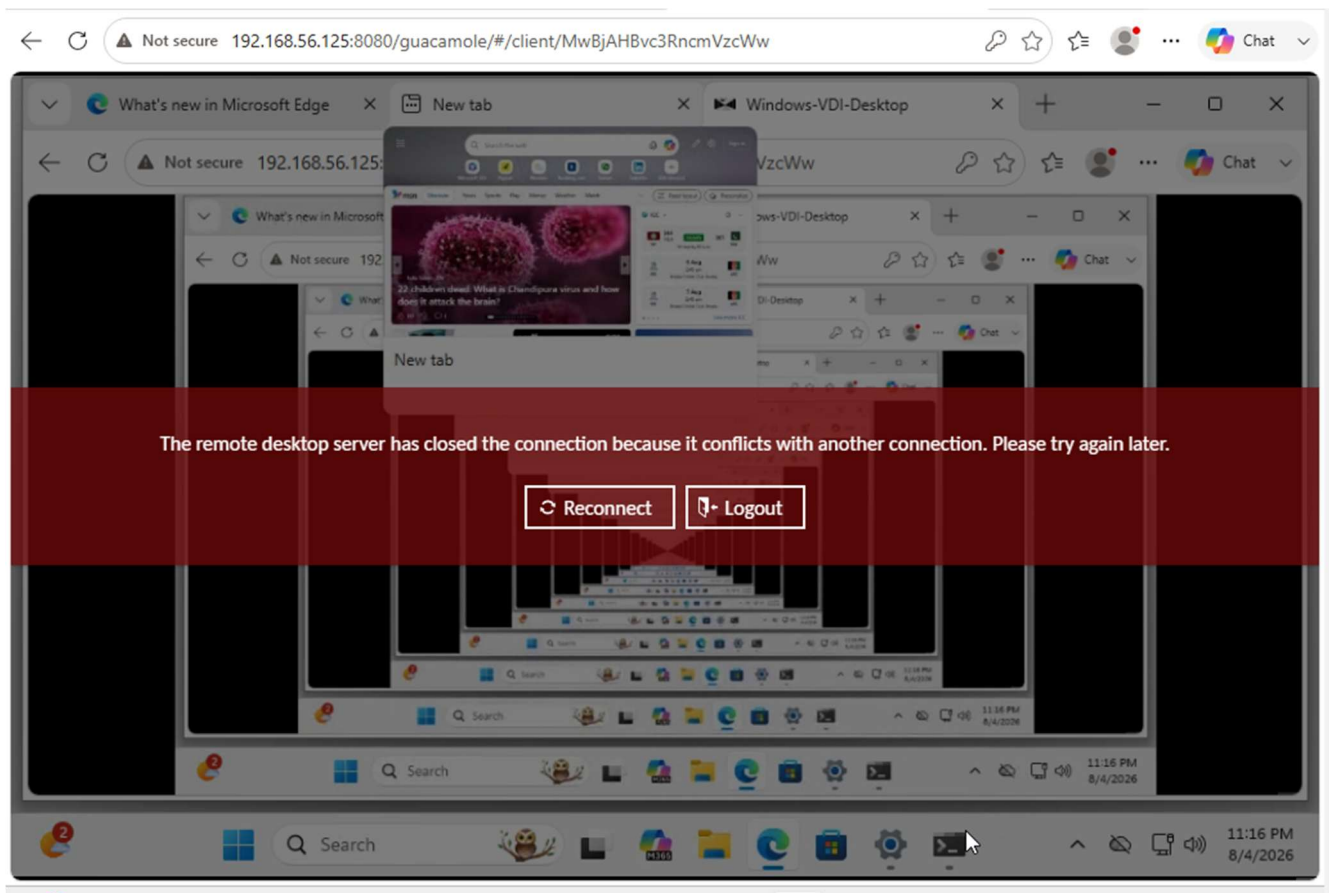
☐

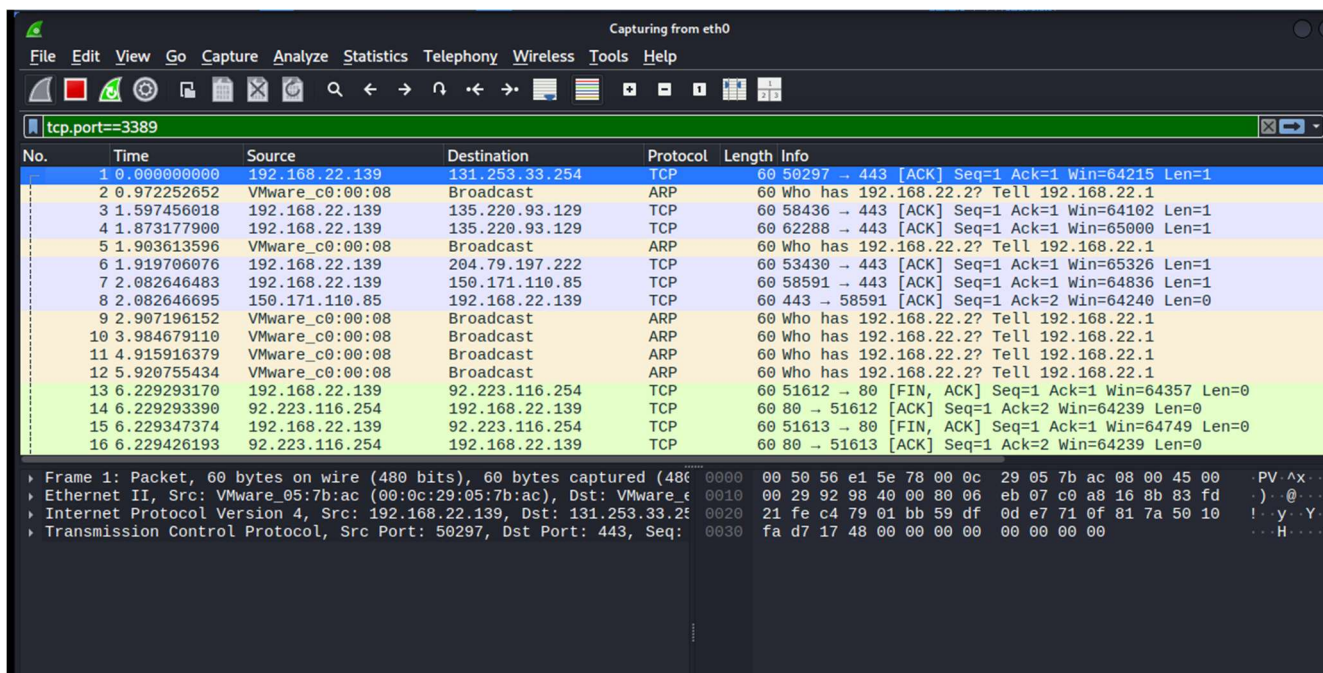
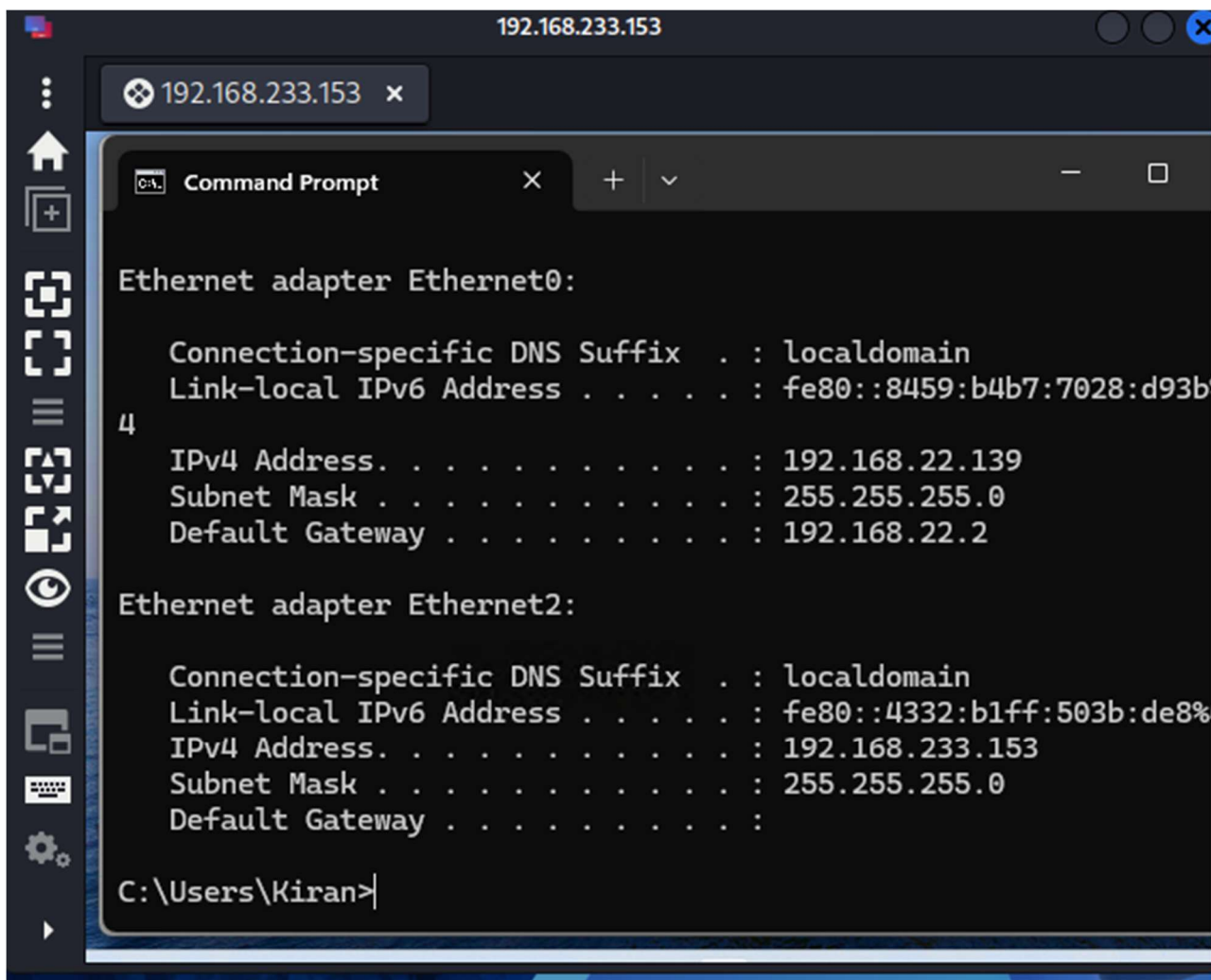
Audit system:

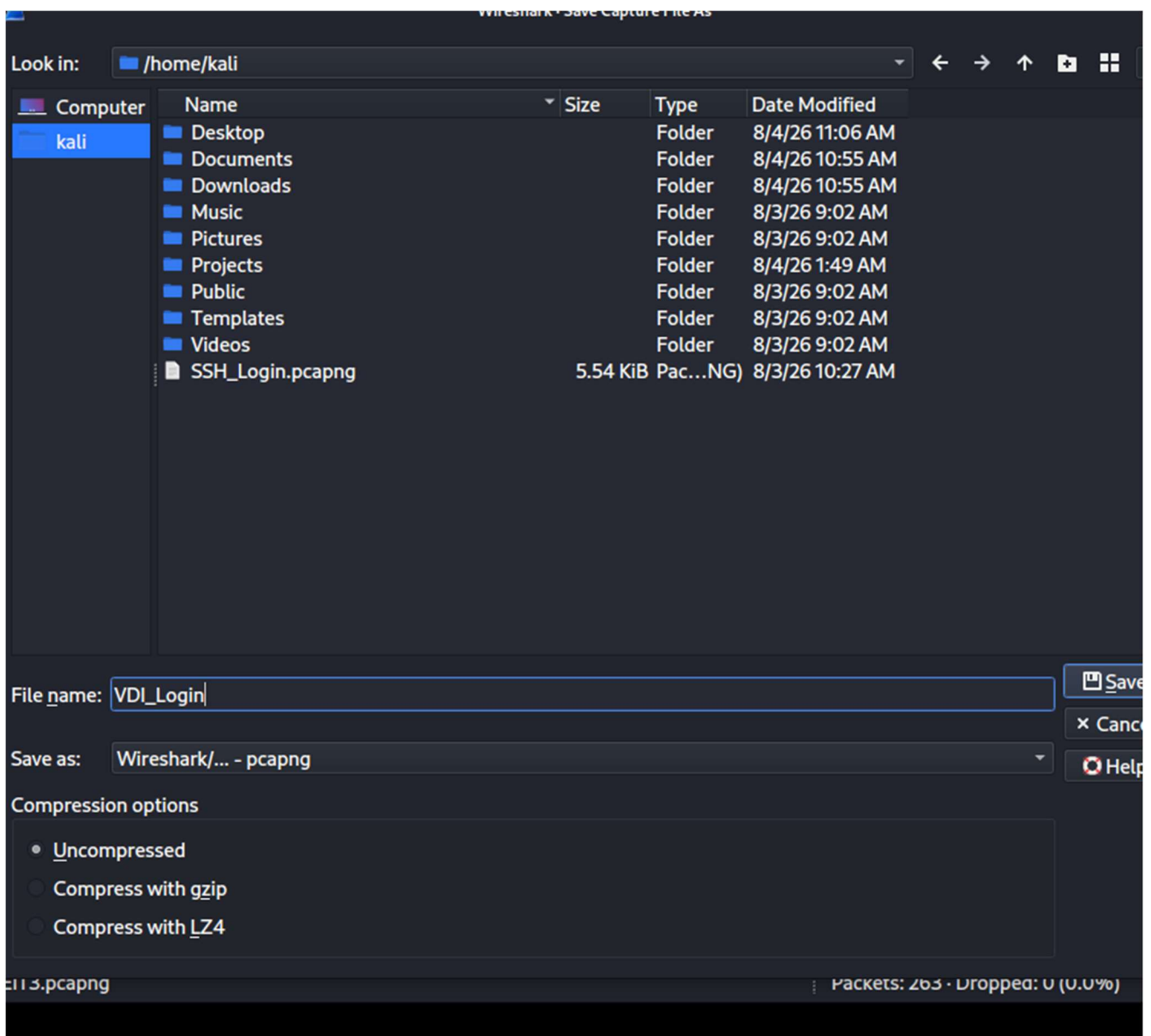
☐

Create new users:

☐







Learning Outcomes

- Gain practical hands-on fluency in executing diverse remote access login methods.
- Identify network traffic signatures for major remote access technologies.

Notes/Observations

SSH connection was established using TCP port 22 between Kali and Ubuntu.

Wireshark showed encrypted SSH traffic, and user credentials/data were not visible.

RDP connection was established using TCP port 3389 to access Windows remotely.

TLS/NLA encryption was observed, with higher bandwidth usage due to graphical desktop transfer.

VPN tunnel was created using UDP port 1194.

Wireshark captured encrypted VPN packets and showed secure tunnel establishment.

VDI session was accessed through a web browser using HTTPS (TCP port 443).

Desktop activities were transmitted securely through encrypted TLS communication.

SSH is best for command-line administration, RDP for remote Windows desktops, VPN for secure network access, and VDI for centralized virtual desktops. Wireshark helped identify protocol traffic and encryption patterns.
